PRODUCT REQUIREMENT DOCUMENT (PRD)

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Project Name: Intelligent E-commerce System - ***EZ System***

Version: 1.0

1. Project Overview

- In Malaysia and globally, small and medium-sized enterprises (SMEs), particularly online sellers on e-commerce, face significant challenges in making timely and informed business decisions. These decisions such as pricing adjustments and inventory planning are often based on fragmented data, limited analytical capabilities, and intuition rather than evidence. As a result, SMEs frequently encounter overstocking and cash flow constraints, underpricing and reduced profit margins, inability to respond to external economic changes such as inflation, supply chain disruptions and so on.

- **Target Domain:** AI-powered Intelligence System E-commerce System, focusing on pricing optimization, inventory decision-making and market-aware business strategy.

- **Proposed Solution Summary:** The EZ System is an AI-powered decision intelligence platform that leverages Z.AI's GLM to help SME owners, particularly e-commerce sellers, make smarter pricing and inventory decisions. It integrates structured business data such as sales, stock levels, and profit margins with unstructured inputs like customer reviews to generate context-aware insights. Rather than simply presenting data, the system interprets business performance, incorporates economic signals such as inflation (CPI/PPI), and recommends actionable strategies including price adjustments, restocking, and operational optimization. By providing clear explanations, trade-off analysis, and measurable impact forecasts, the system enables users to make informed, data-driven decisions that improve profitability and efficiency.

2. Background & Business Objective

- **Background of the Problem:** Many Malaysian SMEs, particularly e-commerce sellers, manage pricing and inventory using intuition or basic spreadsheets. These approaches fail to effectively combine structured business data, such as sales, stock levels, and profit margins with external and unstructured signals like economic changes (e.g., inflation). As a result, decision-making becomes reactive rather than proactive, leading to common issues such as overstocking, missed sales opportunities, and reduced profit margins. This highlights a clear "decision intelligence gap" in everyday business operations.

- **Importance of Solving This Issue:** This system addresses the gap by transforming raw business data into actionable insights through an AI-powered decision intelligence platform. Instead of manually analyzing spreadsheets, users can manage products, monitor performance through a centralized dashboard, and trigger AI analysis to receive pricing, restocking, and optimization recommendations. By shifting from guesswork to data-driven decision-making, SMEs can reduce costs, improve inventory control, and maximize profitability with minimal time and effort.

- **Strategic Fit / Impact:** The system aligns with the AI for Economic Empowerment domain by equipping SMEs with accessible, intelligent decision-making tools. Z.AI's GLM serves as the core engine within the analytics module, interpreting data, generating context-aware recommendations, and explaining decisions in a clear and user-friendly manner. The platform delivers measurable impact through improved revenue, optimized costs, and significant time savings, as demonstrated in the AI impact analysis. By integrating AI directly into daily business workflows, the system enhances productivity and strengthens the economic resilience of local entrepreneurs.

3. Product Purpose

3.1. Main Goal of the System:

To develop an AI-powered decision intelligence system that leverages Z.AI's GLM to transform business data into actionable pricing and inventory decisions for SMEs.

3.2. Intended Users (Target Audience):

- SME business owners, particularly e-commerce sellers
- Online sellers on platforms such as Shopee
- Small retail shop operators managing inventory and pricing
- Entrepreneurs and early-stage business founders
- Users with limited data analysis expertise who require intelligent decision support

4. System Functionalities

4.1. Description

The EZ System functions as a context-aware decision intelligence engine for SME retailers, integrating internal business data such as sales, stock levels, and pricing with external signals like inflation and customer feedback. Powered by Z.AI's GLM, the system analyzes this data to generate actionable recommendations on pricing and inventory, presented through a clear and user-friendly interface. By transforming raw data into explainable insights, the system enables users to make faster and more informed business decisions.

4.2. Key Functionalities:

- **AI-Driven Business Analysis:**
  Processes structured data (sales, stock, pricing) and selected external inputs to generate insights on business performance.

- **Dynamic Inventory Optimization:**
  Evaluates stock levels against demand trends to recommend whether to restock, reduce, or maintain inventory levels.

- **Smart Pricing Recommendations:**
  Suggests price adjustments based on cost changes, demand patterns, and profit optimization goals.

- **Explainable Decision Support:**
  Provides clear, natural language explanations, including reasoning, trade-offs, and expected impact, to help users understand and trust each recommendation.

4.3. AI Model & Prompt Design

4.3.1. **Model Selection:** The system uses Z.AI's GLM as the core large language model due to its strong capability in natural language understanding, structured data interpretation, and reasoning over business-related inputs. These capabilities make it suitable for SME decision intelligence tasks such as pricing optimization, inventory

planning, and trade-off analysis between cost, demand, and profit. GLM is selected because it can transform raw business data into contextual, explainable recommendations, which aligns with the system's goal of supporting data-driven decision-making for SMEs.

4.3.2. **Prompting Strategy:** The system uses a zero-shot prompting approach, where structured business data and clear instructions are directly provided to the model without examples. This approach is suitable because SME data (sales, stock, pricing) is already structured, allowing GLM to efficiently generate insights and recommendations in real time.

4.3.3. **Context & Input Handling**: The system processes structured inputs such as product, sales, and inventory data. Inputs are validated and formatted before being sent to the model. If inputs exceed system limits, they are truncated to prioritize the most recent and relevant data or split into smaller batches. Invalid or unusable inputs are rejected with an error message.

4.3.4. **Fallback & Failure Behavior:** If the model produces irrelevant or unreliable outputs, a basic validation step checks consistency with the input data. The system will retry with a refined prompt or return a fallback message informing the user that analysis cannot be generated, ensuring safe and reliable recommendations.

5. User Stories & Use Cases

- **User Stories:** As an SME e-commerce seller, I want to input or upload my product performance data, including sales volume, total revenue, estimated profit, conversion rate, customer rating, CPI, and PPI, so that the system can automatically process and analyse my business performance in a structured and intelligent way. The system should then combine rule-based business logic with AI-powered reasoning to generate clear insights such as product recommendations, trade-off analysis, impact evaluation, and future trend predictions. This allows me to understand which products are performing well, which ones are underperforming, and what actions I should take to improve profitability without needing to manually calculate or interpret complex business metrics. Ultimately, I want the system to help me make faster, data-driven decisions that improve efficiency, reduce losses, and optimise overall business performance.

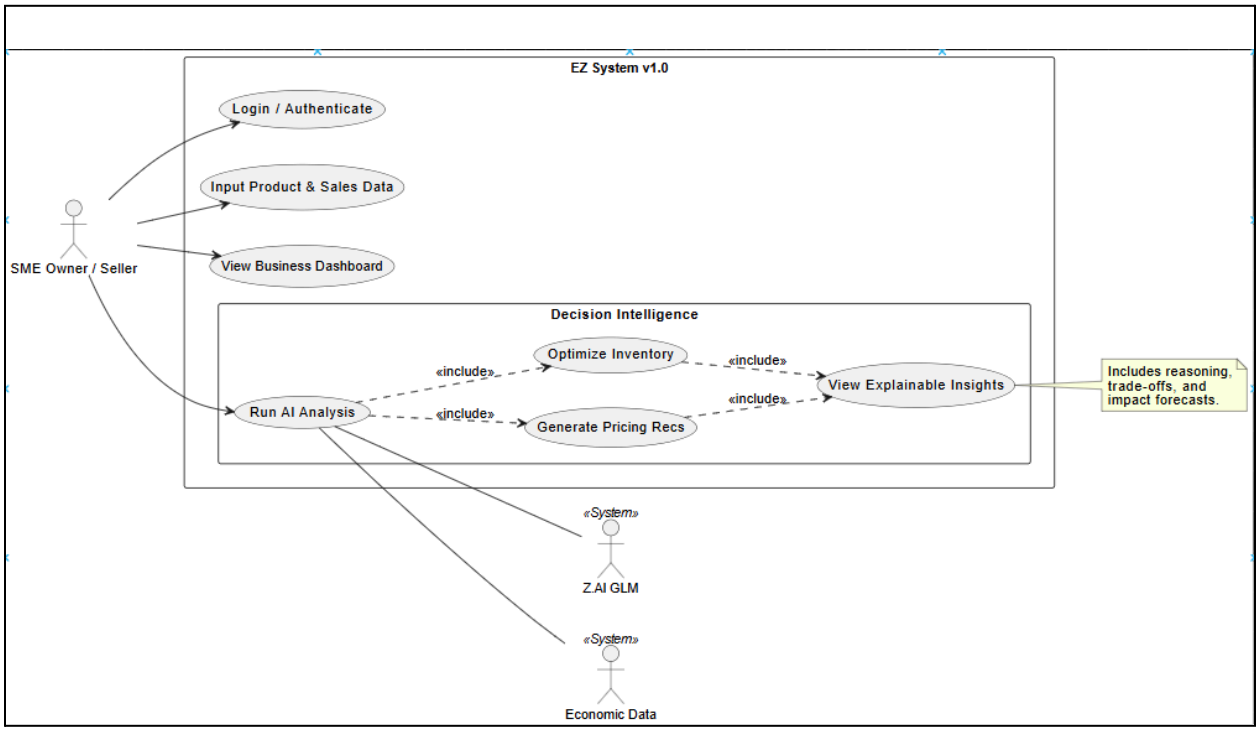
- **Use Cases (Main Interactions) :**

  Prompt to UI Flow:
  Decision Prompt to Insight Flow: Users input structured business data through the Flutter Product Page, including product name, cost price, selling price, stock quantity, and units sold. The system captures this input and sends it to the AI Analytics module when the user triggers the "Run AI Analysis" button. Z.AI's GLM then processes the data to understand business performance, identify patterns, and generate insights. The results are displayed in the Analytics Page as recommendations, forecasts, trade-off analysis, and impact summaries in a structured dashboard format.

  Flow Generation: When the user initiates AI analysis, the system activates the decision intelligence pipeline. GLM analyzes sales, cost, and inventory data to generate actionable outputs such as pricing adjustments, restocking decisions, and profit forecasts. The system then dynamically updates the UI to display AI recommendations, explanations, and impact analysis. This ensures a continuous

flow from data entry → AI reasoning → decision output, enabling real-time business decision support.

- **Use Case Diagram:**



6. Features Included (Scope Definition)

- A Flutter-based SME dashboard that displays business KPIs such as revenue, profit, stock levels, and inventory status.
- Product management system (CRUD) allowing users to add, edit, delete, and view product data including cost, selling price, stock quantity, and units sold.
- AI-powered analytics module that uses Z.AI's GLM to generate pricing recommendations, inventory decisions, forecasts, and trade-off analysis.
- Explainable AI output that provides clear reasoning and impact analysis (revenue, cost, time savings).
- Alert system for business risks such as low stock, profit drop, and demand changes.
- Real-time decision support flow from data input , AI processing and actionable insights.

7. Features Not Included (Scope Control)

- No full external data integration (e.g. live market feeds, competitor APIs, or real-time economic databases).
- No multi-user collaboration or shared business dashboards (single-user SME model only).
- No advanced machine learning training or model fine-tuning (uses pre-trained Z.AI GLM only).
- No complex financial accounting system (e.g. taxes, balance sheets, or auditing features).
- No chatbot-style conversational interface; AI interaction is triggered through structured analysis requests only.

8. Assumptions & Constraints

- **LLM Cost Constraint:** The SME AI system estimates that an average user session consumes around 4,000 to 15,000 tokens, depending on the number of products processed and the KPI data included in the prompt. To control inference cost, the system uses a rule-based preprocessing layer (rule_engine) to generate insights before sending data to the LLM, which reduces prompt size and token usage. In addition, a fallback mechanism is used to return rule-based results when AI processing is unnecessary, minimizing API calls and overall cost.
- **Technical Constraints:** The system is implemented using a Flutter mobile application for the frontend and a Python-based backend for processing business logic and AI integration. The backend is responsible for handling data processing, validation, and communication with the **Z.AI GLM-5.1** for generating business insights and recommendations. The system does not include full enterprise integrations with real e-commerce platforms in the MVP stage. Instead, data is manually input , and all AI interactions are handled through secure API calls between the Python backend and the GLM service.
- **Performance Constraints:** AI response quality depends on input size and clarity. To maintain performance, inputs are structured and limited to key business fields only. Large

or irrelevant inputs are filtered or truncated before being sent to the model to ensure fast and stable response generation.

- **User Input:** The system requires user-provided business data (product details and sales inputs) to generate insights. Without valid input, the AI cannot produce meaningful recommendations, as it relies on structured SME data for analysis.

9. Risks & Questions Throughout Development

- **Data Reliability Risk:** How do we ensure that the system makes accurate decisions when SME input data (sales, inventory, pricing) is incomplete, incorrect, or outdated, and what safeguards are in place to prevent misleading recommendations?
- **AI Hallucination Risk:** What mechanisms are implemented to prevent Z.AI GLM from generating incorrect or overconfident business recommendations, especially when external economic signals are ambiguous or conflicting?
- **Over-Reliance on AI Decisions:** How do we ensure that SME users do not blindly follow AI-generated recommendations without understanding the underlying reasoning, potentially leading to poor financial decisions?
- **Scalability Question:** If multiple users increase system usage, how will API cost and response time be managed efficiently while maintaining performance?